

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \multimap A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \multimap PA$ (8.1).

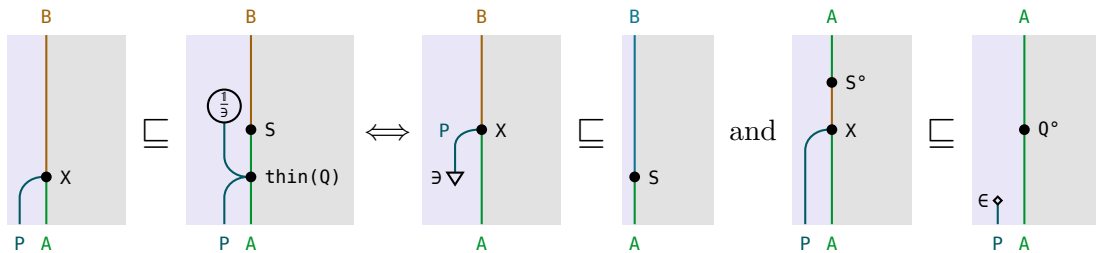
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \multimap ys \multimap xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \multimap b \multimap xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

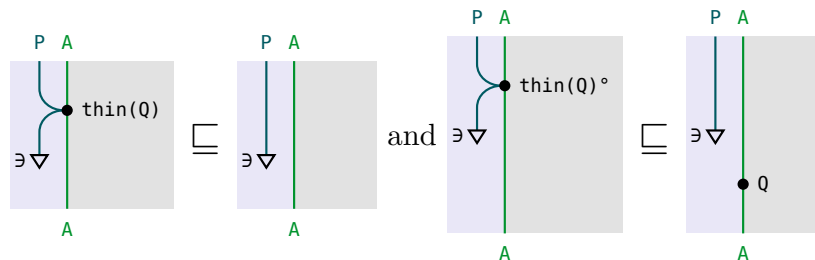
the law	what it says
1 $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$	S is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ni \subseteq \exists$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ni \subseteq \exists \ Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \ Q \subseteq Q \implies \text{thin}(Q) \text{ thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R) \text{ preorder}(Q)$ $\implies \text{est}(R) \subseteq \text{thin}(Q) \text{ est}(R)$	thin-introduction : thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists} \quad (8.2) - (8.2) \text{ est}(Q) \frac{1}{\exists} \subseteq \text{thin}(Q)$	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q) \quad (8.4) - (8.4)$ $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$	thinning each member set is a thinning of the union

§8.1.1. $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



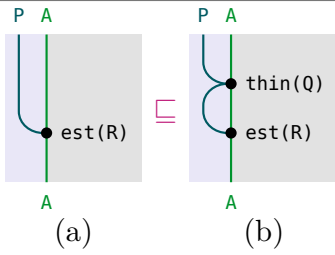
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$ at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\frac{\exists}{\exists} = \Lambda(\exists) = 1$

§8.1.2. $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$\text{preorder}(Q) \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \quad (8.1.2a)$$

est(R)

the $\sqsubseteq$ half: keeping everything is a
thinning — $1 \sqsubseteq Q$



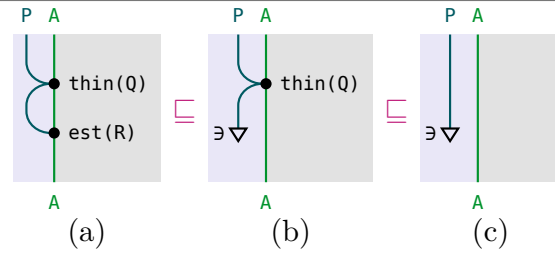
(a)	est(R)	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q),$ $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q)$ $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the $\sqsubseteq$ half, first condition of the UP of

est at $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq$

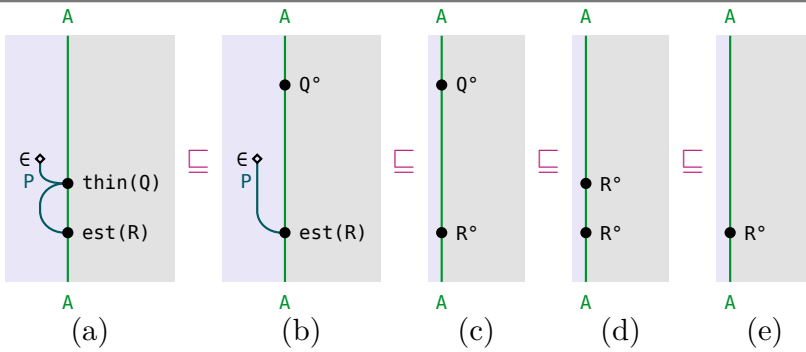
est(R) $\Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



(a)	thin(Q) est(R)	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge \text{preorder}(R) \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition



(a)	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(b)	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	
(e)	$\sqsubseteq R^\circ$	

$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set — $\triangleq \mathbf{thin}, \ni/\ni$ half



(a)	$\mathbf{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$
(b)	$= \mathbf{est}(Q)$	$R \cap S \sqsubseteq R$
(c)	$\sqsubseteq \ni$	

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every member of the set — $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$ half



(a)	$\in \mathbf{est}(Q) \stackrel{1}{\exists}$	$\in \mathbf{est}(R) \sqsubseteq R^\circ$
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \in$
(c)	$\sqsubseteq Q^\circ \in$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

every member of the singleton is a value S returns — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, first condition



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$	$\frac{1}{\exists} \ni = 1$
(b)	$= \frac{S}{\exists} \text{est}(R)$	$R \cap S \subseteq R$
(c)	$\subseteq \frac{S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, second condition



(a)	$S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$
(b)	$= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$T^\circ \frac{1}{\exists} \subseteq \in$
(c)	$\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$\in \text{est}(R) \subseteq R^\circ$
(d)	$\subseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$	$R \cap (S^\circ S) \subseteq Q$
(e)	$\subseteq Q^\circ \frac{1}{\exists}$	$\frac{1}{\exists} \subseteq \in$
(f)	$\subseteq Q^\circ \in$	

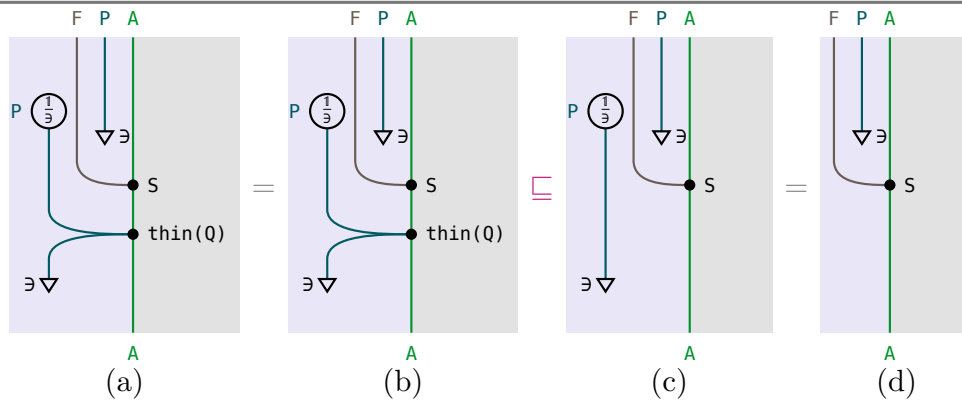
$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

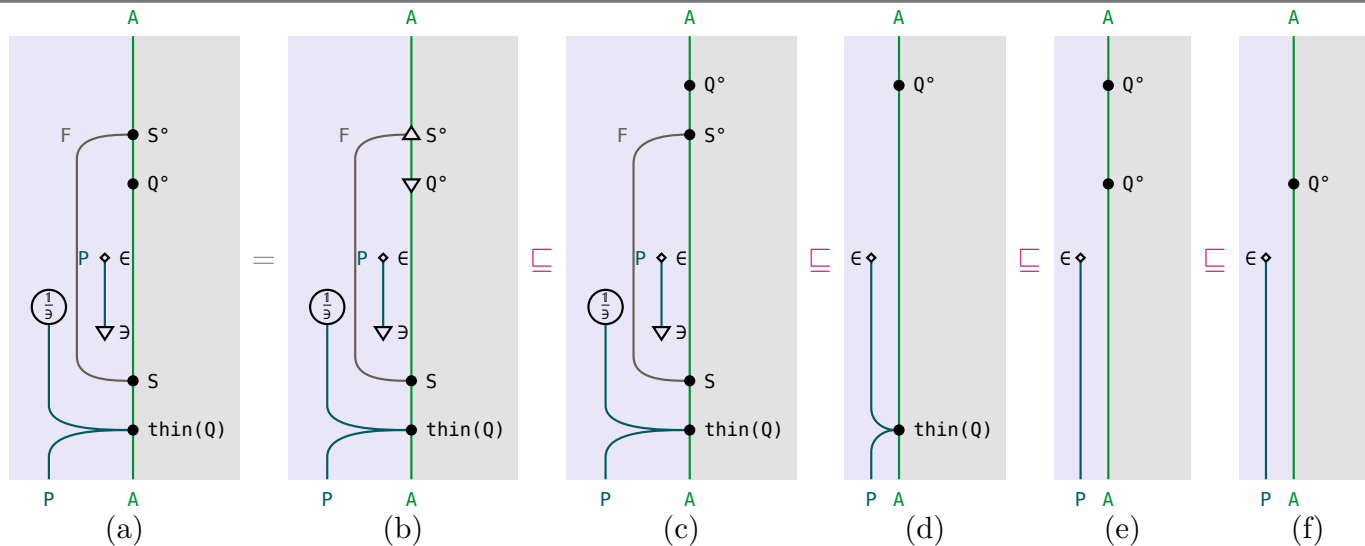
everything the thinning fold keeps is a value of the plain fold — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition, by fusion from the chain below



(a)	$\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	
(b)	$= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	$\text{thin}(Q) \ni \sqsubseteq \exists$
(c)	$\sqsubseteq \frac{F(\exists) S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= F(\exists) S$	

$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is Q -above one the thinning fold keeps — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition, by hylomorphism theorem from the chain below



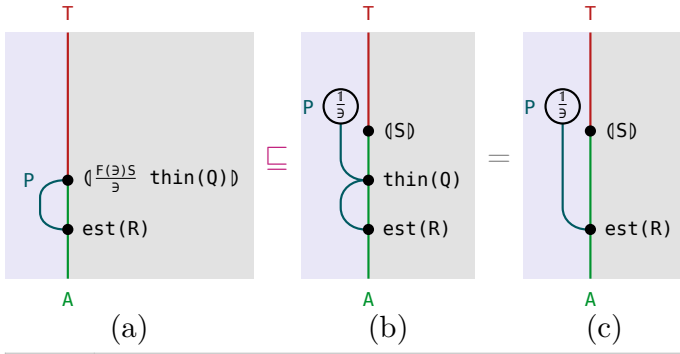
(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	
(b)	$= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$F(Q) S \sqsubseteq S Q$
(c)	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
(d)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$
(e)	$\sqsubseteq Q^\circ Q^\circ \in$	$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$
(f)	$\sqsubseteq Q^\circ \in$	

(8.1.2d)

	step	type	example	what it does
1	$S:F(A)\rightarrow A$ is 8.1's algebra; example $F(X)=V+V\times X$, $S=[wrap,cons]$, candidate $a=cons(v,p)$			
2	S°	$A\rightarrow FA$	$cons(v,p) \mapsto r(v,p)$	takes a apart into the input of its last step
3	$F(Q^\circ\in)$	$FA\rightarrow F(PA)$	$r(v,p) \mapsto r(v,ps)$, some $p'\in ps$ with $p' \ Q \ p$	replaces each recursive component by a set holding a Q -no-worse one; v stays
4	$\frac{F(\exists)S}{\exists}$	$F(PA)\rightarrow PA$	$r(v,ps) \mapsto \{cons(v,t) \mid t\in ps\}$	picks one element from each set, combines by S , collects every result
5	$thin(Q)$	$PA\rightarrow PA$	$\{cons(v,t) \mid t\in ps\} \mapsto ys$	keeps a subset; every dropped element is Q -beaten by a kept one
6	$\sqsubseteq Q^\circ\in$	$A\rightarrow PA$	ys holds y with $y \ Q \ cons(v,p') \ Q \ cons(v,p) = a$	the thinned set still holds an element Q -no-worse than a (S monotonic on Q , Q transitive)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) S \sqsubseteq S Q \Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{S} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



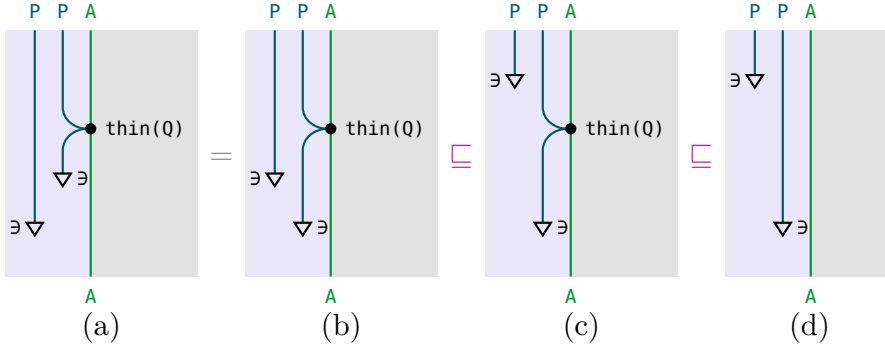
(a)	$(Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R)$	Theorem 8.1 $\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q$ $\Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \sqsubseteq \frac{(S)}{S} \text{thin}(Q)$
(b)	$\sqsubseteq \frac{(S)}{S} \text{thin}(Q) \text{est}(R)$	$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R)$ $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$
(c)	$= \frac{(S)}{S} \text{est}(R)$	

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists \exists$$

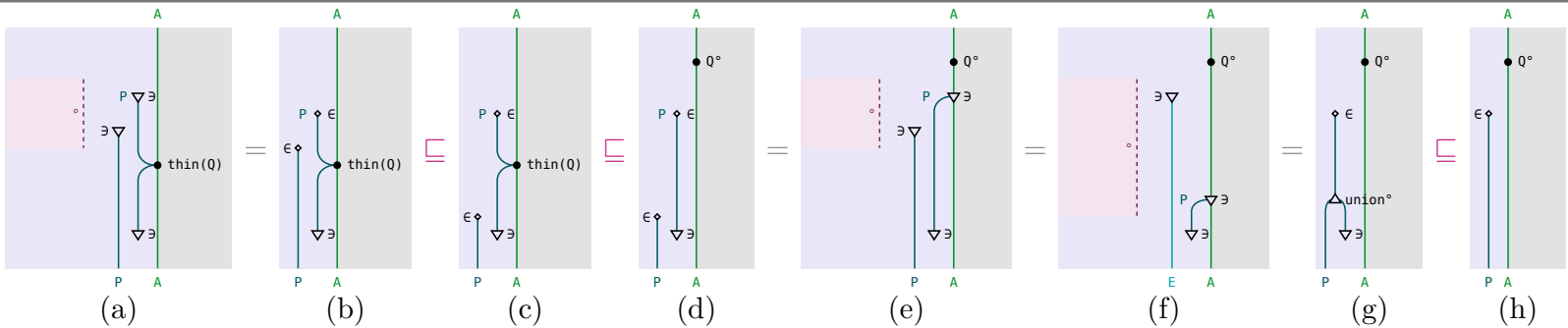
every member of the result is a member of a member set — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition



(a)	$P(\text{thin}(Q)) \text{ union} \exists$	$\text{union} \exists = \exists \exists$
(b)	$= P(\text{thin}(Q)) \exists \exists$	$P(R) \exists \sqsubseteq \exists R$
(c)	$\sqsubseteq \exists \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(d)	$\sqsubseteq \exists \exists$	

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a Q -lower bound in the result — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition



(a)	$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$	$(R S)^\circ = S^\circ R^\circ$
(b)	$= \in \in P(\text{thin}(Q)) \text{ union}$	$\in P(R) \sqsubseteq R \in$
(c)	$\sqsubseteq \in \text{thin}(Q) \in \text{union}$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(d)	$\sqsubseteq Q^\circ \in \in \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= Q^\circ (\exists \exists)^\circ \text{union}$	$\text{union} \exists = \exists \exists$
(f)	$= Q^\circ (\text{union} \exists)^\circ \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(g)	$= Q^\circ \in \text{union}^\circ \text{union}$	$\text{union} = E(\exists)$
(h)	$\sqsubseteq Q^\circ \in$	

§8.2. Paths in a layered network

(8.2a)

	name	type	definition	meaning
1	F	$\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	$F(B, X) = B + (B \times X)$	a path is either one vertex a , or a vertex a followed by a path xs
2	α	$F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$	build a path: wrap one vertex as $[a]$, or put a in front of xs
3	wrapz	$A \rightarrow LA \times \mathbb{N}$	$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$	a path of one vertex, no edge, cost 0
4	consw	$A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$	$(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$	put a in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$
5	cost	$LA \rightarrow \mathbb{N}$	$\text{cost} \triangleq \llbracket [\text{wrapz}, \text{consw}] \rrbracket \pi_2$	the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$
6	R	$LA \rightarrow LA$	$R \triangleq \text{cost} \leq \text{cost}^\circ$	xs is cheaper than ys
7	$L(R)$	$LA \rightarrow LB$	$\llbracket F(\exists, 1) \alpha \rrbracket = L(\exists)$	$L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer
8	minpath	$L(PA) \rightarrow LA$	$\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$	the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	Q	$LA \rightarrow LA$	$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$	like R , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$
10	l	$B \rightarrow B + C$	$a \mapsto l(a)$	mark a value as the left alternative
11	r	$C \rightarrow B + C$	$b \mapsto r(b)$	mark a value as the right alternative
12	cpl	$PB \times A \rightarrow P(B \times A)$	$\text{cpl} \triangleq \text{cp}(1 \times A)$	pair every element of the set with the value beside it
13	cpr	$A \times PB \rightarrow P(A \times B)$	$\text{cpr} \triangleq \text{cp}(A \times 1)$	pair the value beside the set with every element of the set
14	$\frac{F(\exists, 1)}{\exists}$	$F(PB, X) \rightarrow PF(B, X)$	$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	cpl transpose: from layer x and a set of paths ps , choose a vertex $a \in x$
15	$\frac{F(1, \exists)}{\exists}$	$F(B, PX) \rightarrow PF(B, X)$	$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$	cpr transpose: from a vertex a and a set of paths ps , choose a tail $xs \in ps$
16	step	$A \times P(LA) \rightarrow LA$	$\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$	put a in front of every path in ps and keep a cheapest one

(8.2b)

	the law	what it says
1	$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2	$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

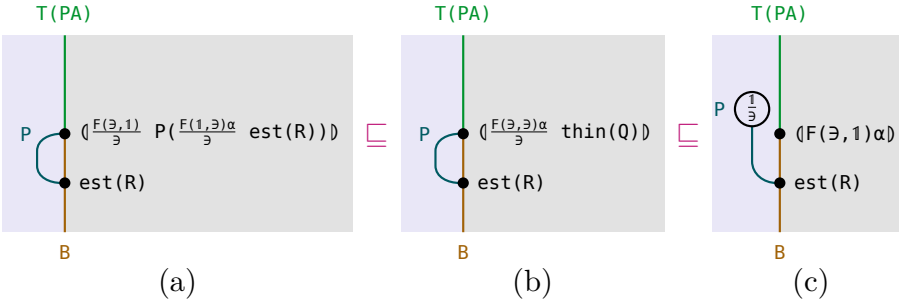
(8.2a)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q) F(\exists, 1) \alpha \sqsubseteq F(\exists, 1) \alpha Q \wedge R \sqcap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \quad (8.2c)$$

$$\sqsubseteq Q \Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

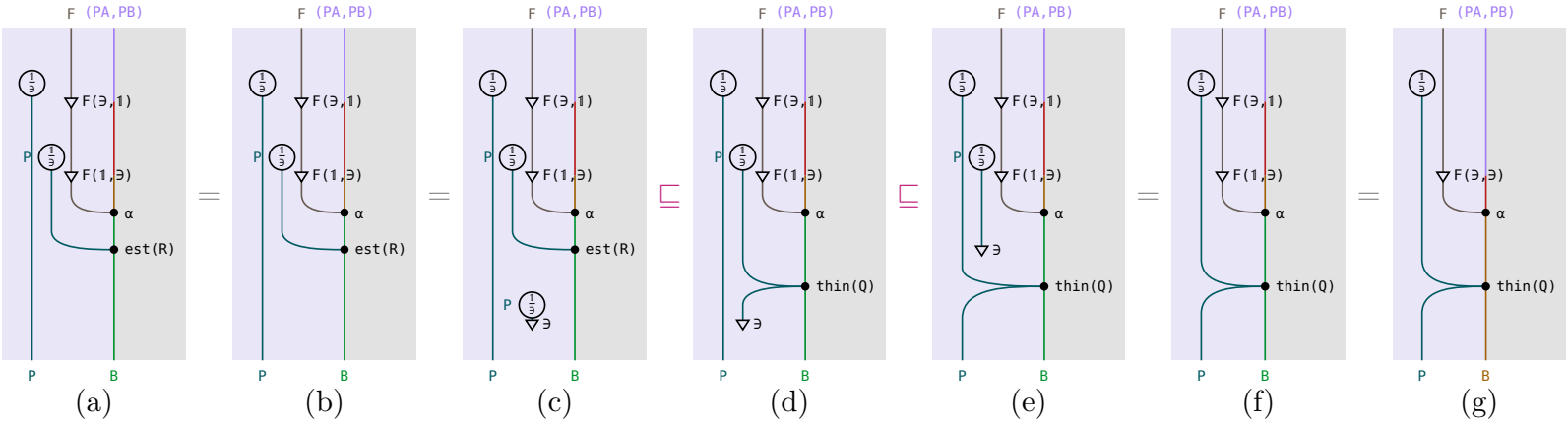
- Why it is needed: $\text{est}(R)$ keeps one cheapest w of the set $\Lambda(S)(x)$; $\text{thin}(Q)$ needs every dropped z to be Q -beaten by a kept one. For $\{w\}$ alone to be a thinning we need $w Q z$ for every z in $\Lambda(S)(x)$.
- We have $w R z$ (w is cheapest) and $w (S^\circ S) z$ (both are built by S from the same argument x); the hypothesis turns the two into $w Q z$: among candidates from one argument, cheaper already means Q -better.



(a)	$\langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R)$	(8.2d) $R \sqcap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$
(b)	$\sqsubseteq \langle \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q)$ $F(\exists, 1) \alpha \sqsubseteq F(\exists, 1) \alpha Q$
(c)	$\sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$	$\Rightarrow \langle \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$

$$S \triangleq F(1, \exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$	
(b)	$= \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$	$P(\frac{1}{\exists}) E(\exists) = 1$
(c)	$= \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \frac{1}{\exists}) E(\exists)$	(8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)$
(d)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) P(\text{thin}(Q)) E(\exists)$	(8.4) $P(\text{thin}(Q)) \text{union} \sqsubseteq \text{union thin}(Q)$
(e)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) E(\exists) \text{thin}(Q)$	$\frac{V S}{\exists} = \frac{V}{\exists} P(\frac{S}{\exists}) \text{union}$
(f)	$= \frac{F(\exists, 1) F(1, \exists) \alpha}{\exists} \text{thin}(Q)$	$F(R, 1) F(1, S) = F(R, S)$
(g)	$= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$
 $P(\text{cons}) \text{est}(R)$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \\ & & R & & S \\ & & \downarrow & & \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and $d R d'$: the leaf moves by R;
- u, v are both inr and $p_1 R q_1 \wedge p_2 S q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr : $R + S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1) = 1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

	name	type	definition	meaning
1	setify	$[A] \rightarrow PA$	$(\text{setify } \exists)(x, a) \iff a \in x$	the set of a list's elements
2	cup	$PA \times PA \rightarrow PA$	$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$	the union of two sets
3	cp(F)	$F(PA) \rightarrow PFA$	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	every F-shaped value whose parts are chosen from the given sets
4	sort($\leq$)	$PA \rightarrow LA$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$	read a set back as one of its $\leq$ -ordered listings
5	bump(Q)	$A \times [A] \rightarrow [A]$	$(a, []) \mapsto [a]$, $\text{bump}(Q)((a, [b] \# xs), ys) \iff Q(a, b) \wedge ys = [a] \# xs \vee Q(b, a) \wedge ys = [b] \# xs \vee \neg Q(a, b) \wedge \neg Q(b, a) \wedge ys = [a] \# [b] \# xs$	put a in front of a thinned list, dropping whichever of a and the old head the other beats
6	thinlist(Q)	$[A] \rightarrow [A]$	$\text{thinlist}(Q) = \emptyset[\text{nil}, \text{bump}(Q)]\emptyset$	thin a list in one pass, bumping each element into the thinned rest
7			$\text{IsThinlist}(Q, \text{thinlist}) \iff \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q)$	what a list implementation of $\text{thin}(Q)$ must do: only drop elements, and thin the set
8	minlist(Q)	$[A] \rightarrow A$	$\text{minlist}(Q) \triangleq \text{setify est}(Q)$	a Q -least element of the list

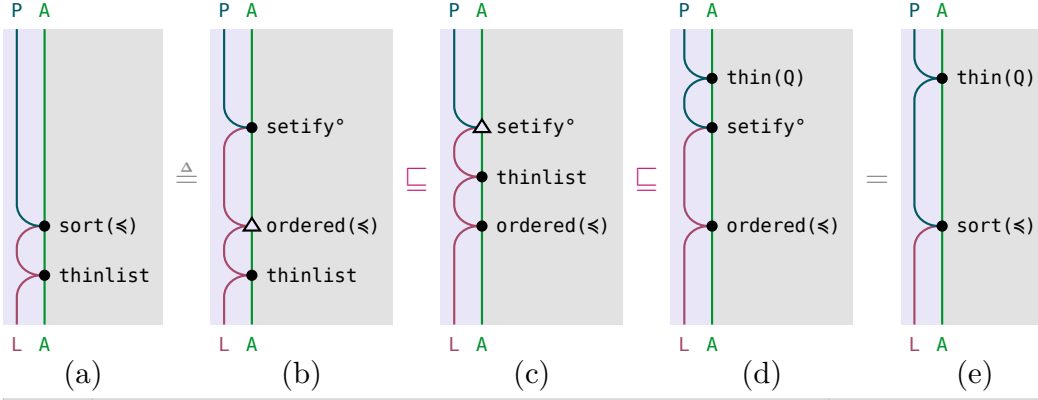
(8.3b)

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$; $\text{listcp} : F(L) \rightarrow LF$.

(8.3c)

	the law	what it says
1	$(\forall a. Q(a, a)) \wedge (\forall a. \forall b. \forall c. Q(a, b) \implies Q(b, c) \implies Q(a, c)) \wedge (\forall a. \forall b. Q(a, b) \vee Q(b, a)) \implies \text{thinlist}(Q) ([a] \# xs, ys) \iff (\exists w. \text{minlist}(Q)([a] \# xs, w) \wedge ys = [w])$ (8.5)	what thinning should come to when it can: one element
2	$\text{IsThinlist}(Q, \text{thinlist}) \implies \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — $(8.6) \text{ Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \implies \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3	$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)	a minimum of the sorted list is a minimum of the set
4	$\text{sort}(g \leq g^\circ) \text{ list}(g) \sqsubseteq P(g) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
5	$\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
6	$(\forall a. \forall b. \forall c. \leq(a, b) \implies \leq(b, c) \implies \leq(a, c)) \wedge (\forall a. \forall b. \leq(a, b) \vee \leq(b, a)) \implies (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

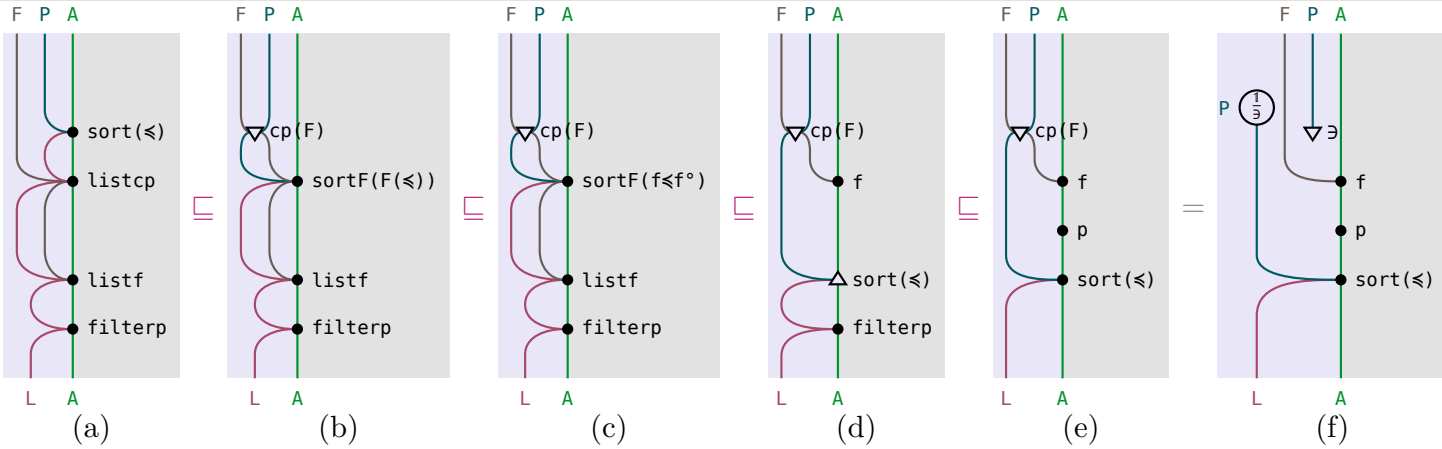
Map(setify) $\wedge$ Coreflexive(ordered($\leq$)) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq ordered($\leq$) $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sort($\leq$) thinlist $\sqsubseteq$ thin(Q) sort($\leq$) (8.6)
 a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, thinlist(Q) $\sqsubseteq$ subseq.



(a)	sort($\leq$) thinlist	sort($\leq$) $\triangleq$ setify $^\circ$ ordered($\leq$)
(b)	$\triangleq$ setify $^\circ$ ordered($\leq$) thinlist	Coreflexive(ordered($\leq$)) $\wedge$ g $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq ordered($\leq$) $\Rightarrow$ setify $^\circ$ ordered($\leq$) g $\sqsubseteq$ setify $^\circ$ g ordered($\leq$)
(c)	$\sqsubseteq$ setify $^\circ$ thinlist ordered($\leq$)	Map(setify) $\wedge$ g setify $\sqsubseteq$ setify T $\Rightarrow$ setify $^\circ$ g ordered($\leq$) $\sqsubseteq$ T setify $^\circ$ ordered($\leq$)
(d)	$\sqsubseteq$ thin(Q) setify $^\circ$ ordered($\leq$)	
(e)	= thin(Q) sort($\leq$)	

**Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow$ sortF(X) $\sqsubseteq$ sortF(Y)) $\wedge$ F($\leq$) f $\sqsubseteq$ f $\leq$ $\wedge$ sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ Lemma 8.1
 P(f) sort($\leq$) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\wedge$ F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$))
 $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$)**

one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA $\rightarrow$ A monotonic
 on $\leq$, p coreflexive, F linear.



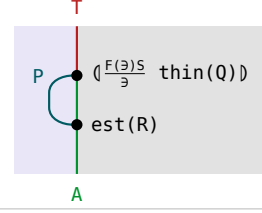
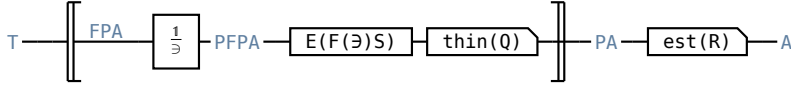
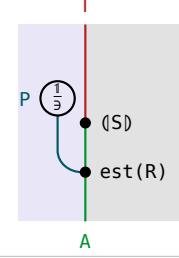
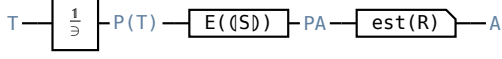
(a)	F(sort($\leq$)) listcp listf filterp	F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$))
(b)	$\sqsubseteq$ cp(F) sortF(F($\leq$)) listf filterp	Map(f) $\Rightarrow$ F(R) f $\sqsubseteq$ f R $\Leftrightarrow$ F(R) $\sqsubseteq$ f R f $^\circ$
(c)	$\sqsubseteq$ cp(F) sortF(f $\leq$ f $^\circ$) listf filterp	sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$)
(d)	$\sqsubseteq$ cp(F) P(f) sort($\leq$) filterp	sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) Map(f) $\Rightarrow$ P(f) = E(f)
(e)	$\sqsubseteq$ cp(F) E(f p) sort($\leq$)	$\frac{\exists}{\exists} E(R) = \frac{\exists R}{\exists}$
(f)	= $\frac{F(\exists) f p}{\exists}$ sort($\leq$)	

Theorem 8.2
 $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad f_1 \ p_1 \sqsubseteq f_1 \ p_1 \ Q \wedge F(Q)$
 $f_2 \ p_2 \sqsubseteq f_2 \ p_2 \ Q \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2$
 $\sqsubseteq f_2 \leq \wedge \text{sortF}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1) \quad \text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2)$
 $\text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_1 \sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge$
 $F(\text{sort}(\leq)) \quad \text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \quad \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge$
 $\text{sort}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \quad \text{filterp}_2 \wedge S = (f_1 \ p_1) \cup (f_2 \ p_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{merge}(\leq))$
 $\text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$

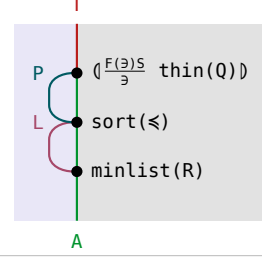
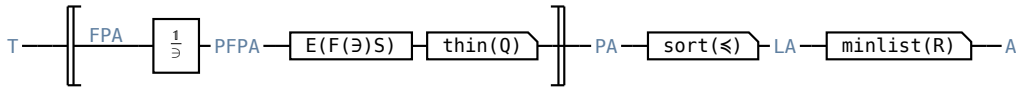
a fold on sorted lists of partial solutions, thinned at every step — at binary thinning data.

circuit

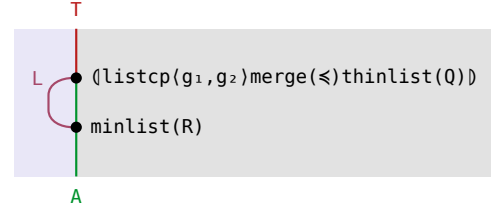
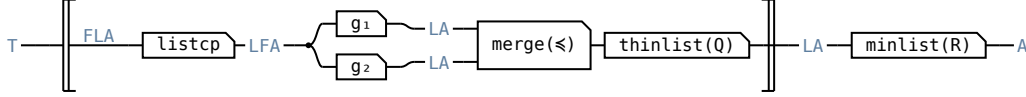
Hinze–Marsden



Corollary 8.1 $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad S \sqsubseteq S \quad Q$
 $\Rightarrow (\frac{F(3)S}{3} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$ at $f_1 p_1$ and $f_2 p_2$ monotonic on Q —
 binary thinning data



$\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R)$ — $\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R)$ at its $Q \sqsubseteq R$



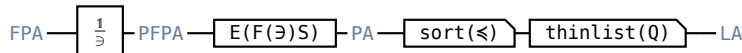
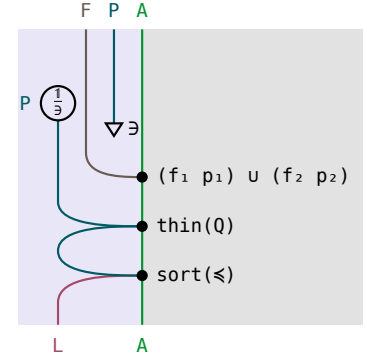
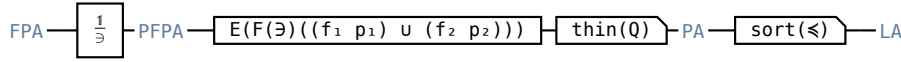
fusion at $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y))$
 $\wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \wedge \text{sortF}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1)$
 $\text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2) \quad \text{sort}(\leq) \wedge \text{sort}(\leq)$
 $\text{filterp}_1 \sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge$
 $F(\text{sort}(\leq)) \quad \text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq))$
 $\text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq)$
 $\Rightarrow F(\text{sort}(\leq)) \quad \text{listcp} \langle \text{listf}_1 \text{filterp}_1, \text{listf}_2 \text{filterp}_2 \rangle \text{merge}(\leq)$
 $\text{thinlist}(Q) \sqsubseteq \frac{F(3) ((f_1 \ p_1) \cup (f_2 \ p_2))}{3} \text{thin}(Q) \text{sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) f_1 \sqsubseteq f_1 \leq \wedge F(\leq) f_2 \sqsubseteq f_2 \leq \quad (8.3d) \\
& \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(\leq) \wedge F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{listcp} (\text{listf}_1 \text{filterp}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\leq) \text{thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$

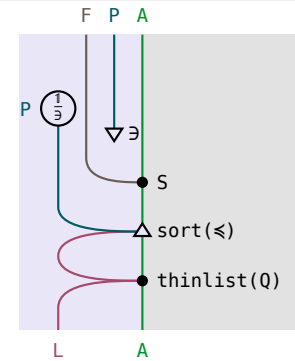
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

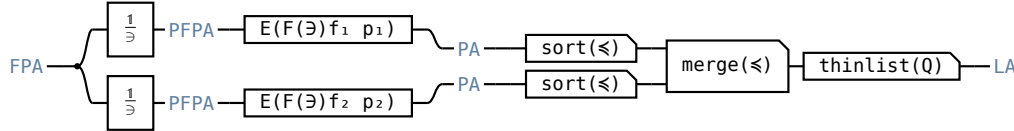
Hinze–Marsden



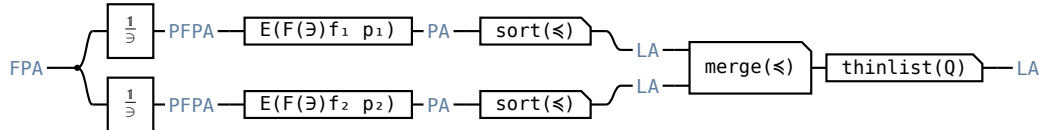
$$\begin{aligned}
& \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq) \text{ — IsThinlist}(Q, \text{thinlist}) \\
& \Rightarrow \text{sort}(\leq) \text{thinlist} \sqsubseteq \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$



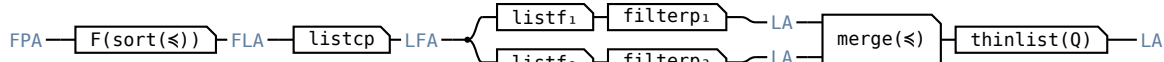
$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \frac{1}{3} \\ \text{PFPA} \rightarrow E(F(\exists) f_1 p_1) \end{array} \right] \text{PA} \rightarrow \text{cup} \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) \text{cup} \text{sort}(\leq) \text{thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) \text{ — binary thinning data, then } \triangleq \text{cup}
\end{aligned}$$



$$\begin{aligned}
& \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \text{thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \text{ — } (\forall a. \forall b. \forall c. \leq(a, b) \Rightarrow \leq(b, c) \Rightarrow \\
& \leq(a, c)) \wedge (\forall a. \forall b. \leq(a, b) \vee \leq(b, a)) \\
& \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq)
\end{aligned}$$



$$\begin{aligned}
& \left(\frac{F(\exists) f_1 p_1}{\exists} \text{sort}(\leq), \frac{F(\exists) f_2 p_2}{\exists} \text{sort}(\leq) \right) \text{merge}(\leq) \text{thinlist}(Q) \\
& (X, Y) (\text{sort}(\leq) \times \text{sort}(\leq)) = (X \text{sort}(\leq), Y \text{sort}(\leq)) \text{ — absorption}
\end{aligned}$$



$$F(\text{sort}(\leq)) \text{listcp} (g_1, g_2) \text{merge}(\leq) \text{thinlist}(Q)$$

Lemma 8.1 $\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge F(\leq) f \sqsubseteq f \leq \wedge$

$$\begin{aligned}
& \text{sortF}(f \leq f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp} \sqsubseteq E(p) \text{sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\leq)) \\
& \Rightarrow F(\text{sort}(\leq)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X(g_1, g_2) \sqsubseteq (Xg_1, Xg_2) \text{ — } X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_i \triangleq \text{list}(f_i) \text{filter}(p_i) \text{ — binary} \\
& \text{thinning data}
\end{aligned}$$

§8.4. The knapsack problem

(8.4b)

(8.4b)

(8.4c)

(8.4b)

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

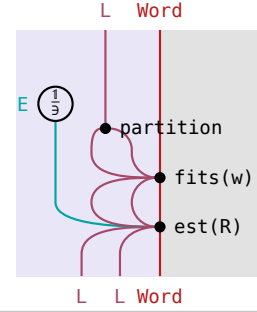
	the law	what it says
1	$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2	$F(Q(\text{len},w)) \text{ [wrap wrap,new]} \sqsubseteq \text{[wrap wrap,new]}$ $Q(\text{len},w)$ $(\forall a. 0 \leq \text{len}(a)) \implies F(Q(\text{len},w)) \text{ [wrap wrap,glue]}$ $\text{ok}(w) \sqsubseteq \text{[wrap wrap,glue]} \text{ ok}(w) \text{ } Q(\text{len},w)$ $\text{cons monotonic on } \text{collect} \leq \text{collect}^\circ.$	both halves are monotonic on Q once ties in waste are broken by the first line
3	$\text{merge}(\tau) = \text{cat}$	τ needs no sorting at all

($\forall a. 0 \leq \text{len}(a) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$ (8.5c)
 $\text{wrap}, \text{new}] \sqsubseteq [\text{wrap wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap wrap}, \text{glue}] \tau$
 $[\text{wrap wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}(w)) \text{sort}(\tau)$
 $\wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup sort}(\tau) \wedge$
 $\text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}$
 $(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\tau) \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\text{est}(R)}$
 a paragraph laid out as a fold that thins the layouts kept at each word

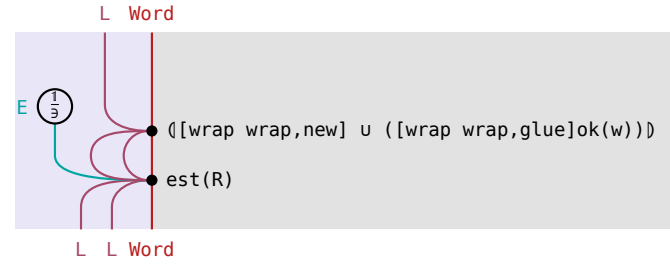
circuit

Hinze–Marsden

$L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E(\text{partition fits}(w))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word}))$
 $\frac{\text{partition list}^*(\text{fits } w)}{\text{est}(R)}$



$L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word}))$
 $= \frac{([\text{wrap wrap}, \text{new}] \sqcup (\text{glue } (\text{ok } w)))}{\text{est}(R)}$
 fusion, every word fits on a line by itself.



$L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word}))$
 $= \frac{([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))}{\text{est}(R)}$
 the algebra as $(f_1 p_1) \sqcup (f_2 p_2)$, $p_1 \triangleq 1$ — binary thinning data.

$L(\text{Word}) \xrightarrow{\frac{1}{3}} P(L(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word}))$
 $([\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)] \text{minlist}(R))$
 binary thinning theorem, at $P \triangleq T$ with $\text{merge } \tau = \text{cat } (\text{merge } \tau) =$
 $\sqsubseteq \text{cat } , Q \text{ from } F(Q(\text{len}, w)) [\text{wrap wrap}, \text{new}] \sqsubseteq [\text{wrap}$
 $\text{wrap}, \text{new}] Q(\text{len}, w), (\forall a. 0 \leq \text{len}(a))$
 $\Rightarrow F(Q(\text{len}, w)) [\text{wrap wrap}, \text{glue}] \text{ok}(w) \sqsubseteq [\text{wrap}$
 $\text{wrap}, \text{glue}] \text{ok}(w) Q(\text{len}, w)$



$= ([\text{start}, \text{cpr } (h_1, h_2) \text{ cat thinlist}(Q)] \text{minlist}(R))$
 $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct — $\triangleq$ partition

§8.6. Bitonic tours

(8.6a)

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$, $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl } \sqcup \text{dropr}])$.

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$, $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $Q \triangleq R \circ (\text{next2 } \text{next2}^\circ) \circ (\text{head2 } \text{head2}^\circ)$, $P \triangleq T$, $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

(8.6b)

the law

what it says

- 1 $(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE
 $(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE

neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists

- 2 $F(Q) [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] Q$
 $F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] Q$,

both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

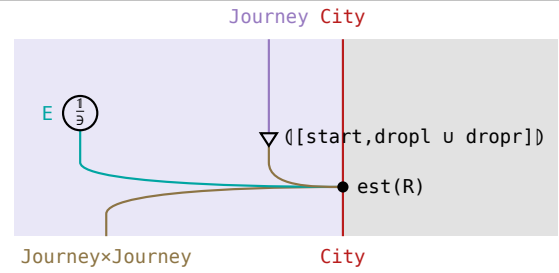
$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}]^\circ) g_1 \sqsubseteq \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{dropr}] \sqcup [\text{start}, \text{dropl}]^\circ) g_2 \sqsubseteq \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \\
& \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow \langle \text{listcp}(g_1, g_2) \text{cat} \\
& \text{thinlist} \rangle \text{minlist} \sqsubseteq \frac{\langle [\text{start}, \text{dropl} \sqcup \text{dropr}] \rangle}{\supset} \text{est}(R)
\end{aligned} \tag{8.6c}$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit

Hinze–Marsden

$$\begin{aligned}
& \mu F \left[\frac{1}{\supset} \right] \text{P}(\mu F) \text{E}(\langle [\text{start}, \text{dropl} \sqcup \text{dropr}] \rangle) \text{P}(\text{Journey}(\text{City})^2) \text{est}(R) \text{Journey}(\text{City}) \\
& \text{tour} \xrightarrow{\supset} \text{est}(R)
\end{aligned}$$

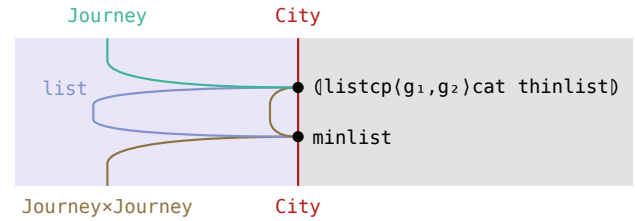


$$\begin{aligned}
& \mu F \left[\frac{1}{\supset} \right] \text{P}(\mu F) \text{E}(\langle [\text{start}, \text{dropl} \sqcup \text{start}, \text{dropr}] \rangle) \text{P}(\text{Journey}(\text{City})^2) \text{est}(R) \text{Journey}(\text{City}) \\
& = \langle [\text{start}, \text{dropl} \sqcup \text{dropr}] \rangle \text{est}(R) \\
& \text{tour} \xrightarrow{\supset} \langle [\text{start}, \text{dropl} \sqcup \text{dropr}] \rangle \text{est}(R) \triangleq \text{tour}
\end{aligned}$$

$$\mu F \left[\frac{1}{\supset} \right] \text{P}(\mu F) \text{E}(\langle [\text{start}, \text{dropl} \sqcup \text{start}, \text{dropr}] \rangle) \text{P}(\text{Journey}(\text{City})^2) \text{est}(R) \text{Journey}(\text{City})$$

$$\langle \text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q) \rangle \text{minlist}(R)$$

binary thinning theorem, at $\text{P} \triangleq \text{T}$ with merge $\tau = \text{cat}$, Q from $F(Q)$
 $[\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \sqcup Q, F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \sqcup Q$



$$\begin{aligned}
& \langle [\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& \text{thinlist}(Q) \rangle \text{minlist}(R) \\
& = \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$